



## Skills

**Mechanical:** SolidWorks (Surfacing, Assemblies), GD&T, DFM/DFA, 3D Printing (Bambu Studio), Mill, Lathe, Rapid Prototyping, Drawings

**Electrical & Hardware:** Altium Designer, Schematic Design, HIL Test Benches, Soldering, Hot Air Rework, Oscilloscope, ESP32, CAN Bus

**Programming:** Python, C++, Embedded C, ROS/DDS, MuJoCo, Sim-to-Real RL, Linux, FreeRTOS, MicroPython, Shell/Bash, Git

**Software & Tools:** Linux, Git, GitLab CI/CD, VS Code, Putty, SolidWorks Simulation, MATLAB, Microsoft Office Suite, Google BigQuery



## Work Experience

### Geotab | Embedded Developer Intern

May 2026 – Aug 2026

*Fleet Management & Telematics*

Oakville, CA

- Integrated 29-DOF Unitree G1 humanoid robot with Inspire dexterous hands for automated warehouse material handling.
- Developed Python GUI with individual joint slider controls and real-time trajectory recorder to automate motion routines.
- Implemented live camera overlay using Google MediaPipe to track hand landmarks and mimic finger postures on Inspire hands.
- Engineered SolidWorks CAD mounts for RealSense depth cameras, touch display, and integrated hardware safety E-Stop circuitry.
- Trained and deployed sim-to-real PPO reinforcement learning policy in MuJoCo for 7-joint arm trajectory tracking over DDS.

### Geotab | System Verification Intern (HIL Team)

Aug 2025 – Dec 2025

*Fleet Management & Telematics*

Oakville, CA

- Developed automated Hardware-in-the-Loop (HIL) test fixtures to validate GNSS, sleep states, and CAN bus vehicle telemetry.
- Integrated prototype telematics units into daily automated HIL test pipelines, automating regression execution via Python scripts.
- Improved automated test suite coverage by 80%, providing critical diagnostic data to expedite embedded firmware bug fixes.

### Geotab | Embedded Developer & QA Intern

Jan 2025 – Apr 2025

*Fleet Management & Telematics*

Oakville, CA

- Gained hands-on production line and warehouse experience assembling tester jigs, flashing PCBs, and performing end-of-line QA tests
- Worked with firmware, mechanical, and hardware teams to debug devices, validate firmware, and configure prototype hardware
- Developed Python and GUI tools to automate device logging and test uploads, improving efficiency for cross-functional teams

### WatArrow & SAE Aero Design East 2025 | Electromechanical Member

Sept 2024 – Apr 2025

*Waterloo Student Design Team & International Competition*

Waterloo, CA

- Prototyped foam control surfaces, manufactured modular balsa wings, and performed carbon fiber layup for custom landing gear
- Wired servo motors to flight computer using modular crimped connections and created detailed electrical diagrams.
- Competed at SAE Aero East 2025; fixed flight computer issue causing flap creep, resulting in 3rd place in mission points.



## Engineering Projects

### 7-DOF QDD Robotic Arm | Personal Project

Jan 2026 – Present

- Engineered a 7-DOF quasi-direct-drive (QDD) robotic arm in SolidWorks achieving a 1:1 payload-to-weight ratio (10 kg payload).
- Designed modular planetary actuators and interlocking spigot joints optimized for PETG 3D printing and M3 heat-set inserts.
- SolidWorks assembly of 375 parts and 110+ fasteners, managing large multipart sub-assemblies with strong DFM intuition.
- Implemented distributed CAN bus architecture with custom ESP32 nodes to execute real-time closed-loop motor trajectory control.

### 10-DOF Robotic Hand | Personal Project

Dec 2025 – Present

- Modeled a 10-DOF robotic hand in SolidWorks, validating grasping kinematics via dynamic mates and adjusting for tolerance.
- Routed high-strength tendon cables through low-friction conduits with spring returns for compliant, multi-object grasping.
- Utilized advanced loft and boundary surfacing techniques to design an ergonomic palm enclosure housing 10 micro-servos.
- Engineered a 10 A transient-safe power rail with 4700  $\mu$ F capacitance and offloaded 50 Hz PWM generation to PCA9685 via I2C.

### Custom PCB Fixture | Internship Project

Mar 2025 – Apr 2025

- Designed a cost effective and modular PCB fixture with an adaptable layout for testing multiple PCB prototypes in lab settings
- SolidWorks assembly of 20+ custom parts and 30 fasteners, integrating a spring-loaded pogo pin plate for rapid PCB mounting.
- Utilized 3D printing for rapid prototyping, tolerance testing, and spring mechanism validation; cut fixture cost by 95%.

### Automatic Door Locker | Personal Project

Sept 2025 – Dec 2025

- Designed a universal door actuator in SolidWorks, simulating gear reduction and tolerancing to maximize dynamic torque output.
- Drafted schematics and routed 2-layer PCB in Altium Designer integrating an LM2596S regulator, INA219 monitor, and TMC2209 driver.
- Built MicroPython web server on RP2040 Pico with sensorless stall detection for auto-calibrating zero-position end stops.

### Airfoil Wind Tunnel | WatArrow Design Team Project

Oct 2024 – Dec 2024

- Designed wind tunnel assembly in SolidWorks with integrated structural mounting for load cell instrumentation and stepper motor.
- 3D printed modular flow nozzles, machined aluminum framing, and routed sensor instrumentation to Raspberry Pi via SPI/I2C.



## Education

### University of Waterloo

Sep 2024 – Present

*Bachelor of Engineering in Mechatronics Engineering*

Waterloo, CA

- Awarded University of Waterloo President's Scholarship of Distinction & Academic Standing: Excellent Standing
- Relevant Coursework:** Structures and Properties of Materials, Mechanics of Deformable Solids 1, Dynamics, Statics, Digital Logic